

Extremal Decomposition of Gibbs Measures with free boundary conditions on Cayley Trees

Arnaud Le Ny⁽¹⁾

⁽¹⁾Université Paris Est Créteil (UPEC)
(Laboratory LAMA - UMR CNRS 8050)

Analysis and Probability Seminar, November 21, 2023
School of Mathematics and Statistics, Wuhan University

Joint work with Loren Coquille (Grenoble) and Christof Külske (Bochum) [1,2]

Program

- 1 Introduction
- 2 Gibbs Measures on Lattice Spin Systems
- 3 Markov Fields on Trees, Markov chains and boundary laws
- 4 Uncountable Family of Inhomogeneous Extremal Gibbs States [1]
- 5 (Continuous) Extremal decomposition of Free States on Trees [2]

Introduction

Context : DLR Gibbs measures for lattice spin systems

Main motivation(s) :

- 1 To compare structure of Gibbs simplicies on trees and on cubic lattices.
- 2 To describe general extremal "glassy states" on trees.
- 3 To describe (continuous) extremal decompositions of "free states".
- 4

Structure of the talk :

- 1 General structure of sets of (DLR) Gibbs measures
- 2 Phase transition and extremal states on $\mathbb{Z}^d$, $d \geq 2$ for *n.n.* Ising models.
- 3 Known extremal states on $\mathcal{T}^d$, $d \geq 2$ for Ising, Potts and SoS models.
- 4 New families and (continuous) extremal decomposition of free states.

Main references used in the talk

[1] **L. Coquille, C. Külske, A. Le Ny.**

Extremal Inhomogeneous Gibbs States for the SoS Model and Finite-Spins Models on Trees. *J. Stat. Phys.* **191**, online n° 71, december 2023.

[2] **L. Coquille, C. Külske, A. Le Ny.** Continuity of the Extremal Decomposition of the Free State for Finite-Spin Models on Cayley Trees.

ArXiv Preprint, submitted. arXiv-2310.1101, october 2023.

[3] **D. Gandolfo, J. Ruiz, S. Shlosman.** A Manifold of Pure Gibbs States of the Ising Model on a Cayley Tree. *J. Stat. Phys.* **148**:999–1005, 2012.

[4] **D. Gandolfo, C. Maes, J. Ruiz, S. Shlosman.** Glassy States : the Free Ising Model on a Tree. *J. Stat. Phys.* **180**:227–237, 2020.

[5] **H.-O. Georgii.** Gibbs Measures and Phase Transitions. *de Gruyter*, Chapters 7 & 12, 1988 (2^d edition 2011).

Program

- 1 Introduction
- 2 Gibbs Measures on Lattice Spin Systems
- 3 Markov Fields on Trees, Markov chains and boundary laws
- 4 Uncountable Family of Inhomogeneous Extremal Gibbs States [1]
- 5 (Continuous) Extremal decomposition of Free States on Trees [2]

Gibbs Measures for Lattice Spin Systems

DLR approach: Equilibrium states at infinite volume

Discrete Lattices S , finite subsets $\mathcal{S} = \{\Lambda \in S, |\Lambda| < \infty\}$

"Usually" $S = \mathbb{Z}^d$, $d \geq 1$. Here $S = \mathcal{T}^d$ or $\mathcal{T}_0^d$ (rooted), $d \geq 2$

Finite single spin state-space $(E, \mathcal{E}, \rho_0)$

- Ising spins : $E = \{-1, +1\}$
- Potts spins : $E = \{1, 2, \dots, q\}$
- SoS Spins : $E = \mathbb{Z}$ (unbounded – *partial results only*)

Infinite configuration space $(\Omega, \mathcal{F}, \rho) = (E^S, \mathcal{E}^{\otimes S}, \rho_0^{\otimes S})$

- **Microscopic states** : Configurations $\sigma = (\sigma_i)_{i \in S}$, $\sigma_i \in E$, for any $i \in S$
- **Macroscopic states** : Probability measures μ on the space $\Omega = E^S$

Interactions : Families $\Phi = (\Phi_A)_{A \in \mathcal{S}}$ s.t. Φ_A is $\mathcal{F}_A$ -measurable.

Hamiltonian with free boundary condition ($\Lambda \in \mathcal{S}$) at finite volume $\Lambda \in \mathcal{S}$:

$$H_\Lambda^\emptyset(\sigma) = \sum_{A \subset \Lambda} \Phi_A(\sigma)$$

Hamiltonian with b.c. $\omega \in \Omega$, finite volume $\Lambda \in \mathcal{S}$:

$$H_\Lambda(\sigma|\omega) = \sum_{A \cap \Lambda \neq \emptyset} \Phi_A(\sigma_\Lambda \omega_{\Lambda^c})$$

Ferromagnetic *n.n.* potentials Φ (positive couplings J_{ij} or $u_{k,l}$)

- Ising model(s) : $\Phi_{\{i,j\}}(\sigma) = -J_{ij}\sigma_i\sigma_j$ or $\Phi_{\{i,j\}}(\sigma) = 1 - \sigma_i\sigma_j$
- Potts model : $\Phi_{\{i,j\}}(\sigma) = \mathbf{1}_{\sigma_i \neq \sigma_j}, \forall i \sim j \in \{1, \dots, q\}$
- Clock models : $\Phi_{\{i,j\}}(\sigma) = \sum_{k,l=1}^q u_{k,l} \cdot \mathbf{1}_{(\sigma_i, \sigma_j) = (k,l)}, u_{k,l} \geq 0$
- p -SoS models : $\Phi_{\{i,j\}}(\sigma) = |\sigma_i - \sigma_j|^p, p > 0$

DLR approach – Regular systems of conditional probabilities

Aim 1 : To maximize free energy "F=U-TS" (2^d law) : " $\mu = e^{-\beta H}$ " ($\beta = \frac{1}{T}$).

Aim 2 : Phase transitions *via* different (extremal) elements of $\mathcal{G}(\gamma)$.

Gibbs specification $\gamma^{\beta\Phi} = (\gamma_{\Lambda}^{\beta\Phi})_{\Lambda \in \mathcal{S}}$, inverse temperature $\beta > 0$

$$\forall \Lambda \in \mathcal{S}, \forall \sigma, \omega \in \Omega, \gamma_{\Lambda}^{\beta\Phi}(\sigma|\omega) = \frac{1}{Z_{\Lambda}^{\omega}} e^{-\beta H_{\Lambda}(\sigma|\omega)}$$

DLR equations for Gibbs measures : convex set $\mathcal{G}(\gamma) \subset \mathcal{M}_1^+(\Omega)$

$$\mu \in \mathcal{G}(\gamma) \iff \forall \Lambda \in \mathcal{S}, \mu[\sigma_{\Lambda} | \mathcal{F}_{\Lambda^c}](\cdot) = \gamma_{\Lambda}(\sigma|\cdot), \mu - \text{a.s.}$$

Quick form : invariant measure of the specification γ

$$\mu \in \mathcal{G}(\gamma) \iff \mu = \mu \gamma_{\Lambda}, \forall \Lambda \in \mathcal{S}$$

Physical States : Extremal Gibbs measures (set $\text{ex } \mathcal{G}(\gamma)$)

Key Property : $\mathcal{G}(\gamma)$ is a Choquet simplex [5]

Any Gibbs measure $\mu \in \mathcal{G}(\gamma)$ can be uniquely decomposed into

$$\mu(\cdot) = \int_{\text{ex } \mathcal{G}(\gamma)} \nu(\cdot) \alpha_{\mu}(d\nu)$$

with a unique **decomposition measure** α_{μ} on $\mathcal{M}_1^+(\text{ex } \mathcal{G}(\gamma))$.

Important goals in Mathematical Statistical Mechanics :

- ① To characterize $\text{ex } \mathcal{G}(\gamma)$ (reference [1] on Cayley trees)].
- ② To describe support and properties of α_{μ} (reference [2] on Cayley trees)

Important **tail σ -algebra** of asymptotic events :

$$\mathcal{F}_{\infty} := \bigcap_{\Lambda \in \mathcal{I}} \mathcal{F}_{\Lambda^c}$$

Peierls argument and interface states for Ising models

Contour representation : $\omega \longleftrightarrow \Gamma = \{\gamma_1, \dots, \gamma_n\}$ at finite volume (1 to 1)

On Ising's model of ferromagnetism

479

There are closed boundaries and open ones, the latter starting and ending on the edges of the array. We show that at sufficiently low temperatures the area enclosed by closed boundaries and cut off by open ones is only a small fraction of the total area. Hence the majority of the magnets must be of equal sign and the model corresponds to our definition of a ferromagnet.

We give an upper limit for the number of boundary lines passing through any one point, assuming statistical equilibrium. Each boundary that starts at a given point may be described as a succession of steps of equal length, of which each may have the direction $\pm x, \pm y, \pm z$ or $\pm w$. Show each element of the

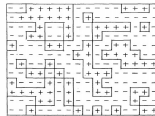


Fig. 1. Example of boundary lines

boundary separates two magnets of opposite sign, the boundary has an energy $2J$, if L is its length, its units of the distance between neighbouring magnets.

However, not every succession of steps represents a possible boundary. Boundaries are restricted by the conditions that

- (1) A boundary must either return to its starting point, or end on an edge.
- (2) It passes through none of its points more than once.
- (3) Not more than two boundaries can pass through any one point and, if there are two, the two steps adjoining the point on one boundary must be different from those on the other.
- (4) No part of a boundary can lie outside the array of magnets.

Figure: Original Peierls Contours (1936)

Phase transition at low Temperature on $\mathbb{Z}^2$: Peierls argument

Energy estimate : cost of a contour and low T stability

- Start from $+$ ground state and insert a defect of $-$'s : contour γ .
- Energetic cost of inserting γ : $H_{\Lambda}^{+}(+) - H(-_{\gamma} +_{\gamma^c}) = 2|\gamma| \implies$ the probability of occurrence of a contour is depressed by $e^{-2\beta|\gamma|}$
- Partition function can be re-written (low T contour representation) :

$$Z_{\Lambda}^{+} = \sum_{\omega \in \Omega_{\Lambda}^{+}} e^{-2\beta \sum_{\gamma \in \Gamma(\omega)} |\gamma|} = \sum_{\omega \in \Omega_{\Lambda}^{+}} \prod_{\gamma \in \Gamma(\omega)} e^{-2\beta|\gamma|}$$

Sketch of Peierls argument of spontaneous magnetization

- The event $\{\sigma_0 = -1\}$ implies the existence of a contour γ^* containing 0
- $\mu_{\Lambda}^{+}(\Gamma \ni \gamma^*) = e^{-2\beta|\gamma^*|} \frac{\sum_{\omega: \Gamma(\omega) \ni \gamma^*} \prod_{\gamma \in \Gamma(\omega) \setminus \gamma^*} e^{-2\beta|\gamma|}}{\sum_{\omega} \prod_{\gamma \in \Gamma(\omega)} e^{-2\beta|\gamma|}} \leq e^{-2\beta|\gamma^*|}$ (spinflip + symmetry)
- low temperature : $\mu^{+}\{\sigma_0 = -1\} \leq \frac{2}{3} \sum_{k \geq 4} k 3^k e^{-2\beta k} \xrightarrow{\beta \rightarrow \infty} 0$

2d pictures : Phase transition at low T vs. Uniqueness

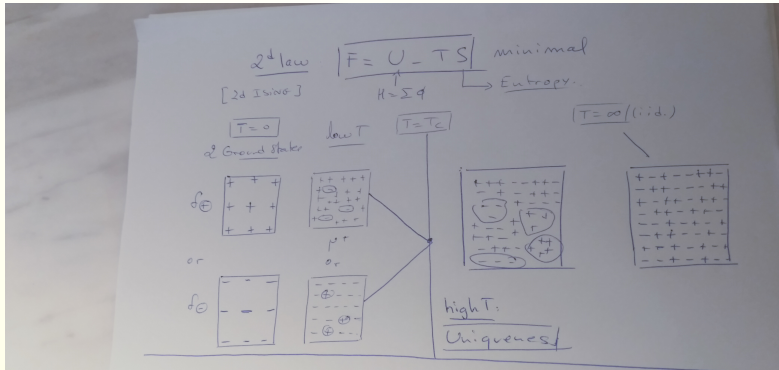


Figure: Sketch of the Temperature driven Phase Transition for 2d Ising model

Fluctuation of interface in $\mathbb{Z}^2$

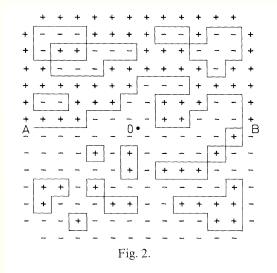


Figure: Interface line in $\mathbb{Z}^2$

- The Gallavotti line AB fluctuates as $\sqrt{L} : \mu^\pm = \frac{1}{2}\mu^- + \frac{1}{2}\mu^+$.
- Aizenman-Higuchi 79-80 : no other extremal states than μ^- and μ^+ .
(In particular, $\mu^f = \frac{1}{2}\mu^- + \frac{1}{2}\mu^+$)

Dobrushin "interface" states in $\mathbb{Z}^3$

Dobrushin 1972 : non-translation invariant extremal state $\mu^\pm \neq \frac{1}{2}\mu^- + \frac{1}{2}\mu^+$.

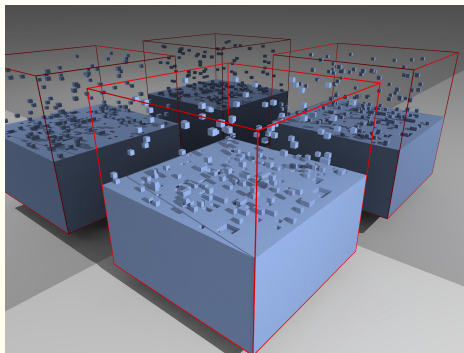


Figure: Dobrushin interface in $\mathbb{Z}^3$

Program

- 1 Introduction
- 2 Gibbs Measures on Lattice Spin Systems
- 3 Markov Fields on Trees, Markov chains and boundary laws
- 4 Uncountable Family of Inhomogeneous Extremal Gibbs States [1]
- 5 (Continuous) Extremal decomposition of Free States on Trees [2]

On trees, **Extremal** Gibbs measures for Markov fields are Markov chains

$n.n.$ Interactions : **Markov fields** (*tools* : Markov chains and boundary laws).

Markov fields on trees [5]

A probability measure on Ω is said to be a **Markov field** iff

$$\forall A \in \mathcal{F}, \forall \Lambda \in \mathcal{S}, \mu[A \mid \mathcal{F}_{\Lambda^c}](\cdot) = \mu[A \mid \mathcal{F}_{\partial\Lambda}](\cdot), \quad \mu - \text{a.s.}$$

Markov chains on trees [5]

A probability measure μ on Ω is said to be a **Markov chain** iff $\forall i < j$ $n.n.$,

$$\forall y \in \mathcal{S}, \mu[\sigma_j = y \mid \mathcal{F}_{1-\infty, ij}](\cdot) = \mu[\sigma_j = y \mid \mathcal{F}_{\{i\}}](\cdot), \quad \mu - \text{a.s.}$$

Important properties [5]

- Gibbs measures with $n.n.$ interactions are Markov fields
- There exist Markov fields which are not Markov chains (Higuchi 77)
- Extremal Gibbs measures for a $n.n.$ potential are Markov chains
- There exist Markov chains that are not extremal (free states !)

Transfer operators and boundary laws on trees

Transfer operator and specification [5]

For any $n.n.$ potential Φ , define $Q : \mathbb{Z}_q \times \mathbb{Z}_q \mapsto (0, \infty)$ for $b = \langle ij \rangle$ as

$$Q(\omega_i, \omega_j) =: Q_b(\omega) = e^{-\beta(\Phi_b(\omega) + (\Phi_i(\omega) + \Phi_j(\omega))/(k+1))}.$$

so that for all $\omega \in \Omega$ and $\Lambda \in \mathcal{S}$, $\gamma_\Lambda(\omega_\Lambda \mid \omega_{\Lambda^c}) = \frac{1}{Z_\Lambda^\omega} \prod_{i \sim j \cap \Lambda \neq \emptyset} Q(\omega_i, \omega_j)$.

Boundary laws and transition matrix [5]

A boundary law $\lambda \in]0, \infty[^{\mathbb{Z}_q}$ for Q is s.t. for all pair $i \sim j \in \Lambda$, $\exists c_{ij} > 0$ s.t.

$$\forall x \in \mathbb{Z}_q, \lambda_{ij}(x) = c_{ij} \prod_{k \in \partial\{i\} \setminus \{j\}} \sum_{y \in \mathbb{Z}_q} Q(x, y) \lambda_{ki}(y).$$

There is a one-to-one relation between $\mu \in \mathcal{G}(\gamma)$ and λ via the marginals

$$\mu(\omega_{\Lambda \cup \partial\Lambda}) = (Z_\Lambda(\omega))^{-1} \prod_{j \in \partial\Lambda} \lambda_{j\Lambda}(\omega_j) \prod_{b \cap \Lambda \neq \emptyset} Q_b(\omega), \quad P_{ij}(\omega_i, \omega_j) = \frac{Q(\omega_i, \omega_j) \lambda_{ji}(\omega_j)}{\sum_{z \in \mathbb{Z}_q} Q(\omega_i, z) \lambda_{ji}(z)}$$

Homogeneous boundary laws in the Ising and Potts cases

Bléher et al. 1991-1995-1998, Ghanikhodaev, Rozikov et al., Kuelske et al., etc.

Homogeneous cases : Boundary laws can be written as $\lambda = e^h$

Consistent families of Gibbs measures with boundary fields

Family $(\mu_\Lambda^h)_{\Lambda \in \mathcal{J}}$ of finite-vol. measures $\mu_\Lambda^h(\sigma_\Lambda) = \frac{1}{Z_\Lambda^h} e^{-H_\Lambda^h(\sigma) + \sum_{x \in \partial \Lambda} h_x \sigma_x}$

Splitting Gibbs Measures (Rozikov *et al.*)

- Infinite-volume Gibbs measure μ^h such that Kolmogorov consistency holds.
- Boundary fields should satisfy the recursion equations : for all $x \in S$

$$h_x = \sum_{j=1}^d f(h_{y_j}, \beta), \quad f(x, \beta) = d \operatorname{argth}(\tanh \beta \tanh x)$$

$\Rightarrow$ 1 or 3 solutions $-h^*, 0, +h^*$ giving rise to μ^-, μ, μ^+ at low temperature

$\Rightarrow$ Interface states by playing on $\pm h^*$ ('Central states', Potts models).

Previous results : extremal interface states on trees

Ising model(s) results

- Phase transition : Preston 1974, Spitzer 1975, Moore-Snell 1979.
- Interface states : Higuchi, Bleher-Ganikhodaev 1991 etc.
- Periodic and weakly periodic states : Rozikov *et al.* 2002–...
- Potts : Ghanikhodaev 1974, Külske-Rozikov 2014, etc.
- Many others by playing on non-linear dynamical systems.

A manifold of extremal Gibbs states ?

Other techniques : Gandolfo *et al.* [3] describe a manifold of extremal states seen as low T perturbations of *sparse ground states* in the Ising case.

Motivation of our work [1]: to extend them to more general Potts, clock or even p -SoS models (by different techniques and proofs, described now). They provide the flavor of the extremal states entering in the decomposition of the free state(s) at low temperature (below the spin-glass transition) [2].

Program

- 1 Introduction
- 2 Gibbs Measures on Lattice Spin Systems
- 3 Markov Fields on Trees, Markov chains and boundary laws
- 4 Uncountable Family of Inhomogeneous Extremal Gibbs States [1]
- 5 (Continuous) Extremal decomposition of Free States on Trees [2]

Sparse inhomogeneous Ground States (Potts model)

Define $d_D := \max_{i \sim j} d_D(j)$ where $d_D(j)$ is the number of bonds incident to j .

Set of broken bonds – Inhomogeneous GS configuration ω^0

A configuration ω^0 is said to be a **sparse inhomogeneous ground set** if its set of broken bonds

$$D := D(\omega^0) := \{\{i, j\} \in S : \omega_i^0 \neq \omega_j^0\}$$

is such that $d_D \leq d_{\max}$, with $d_{\max} < d$ *small enough*.

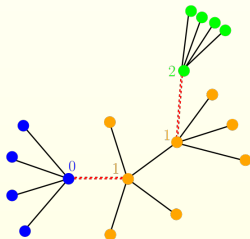


Figure: Ground state configuration ω^0 with $d_{\max} = 1$ for the 3-Potts model.

Sparse inhomogeneous Ground States (SoS and Ising models)

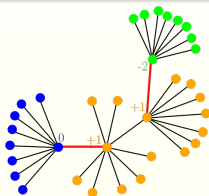


Figure: Ground state configuration ω^0 with $d_{\max} = 1$ for the SoS model.

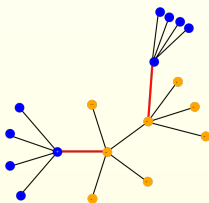


Figure: Ground state configuration ω^0 with $d_{\max} = 1$ for the Ising model.

Contours in a GS configuration ω^0 for Potts or SoS on trees

Labelled contours $\bar{\gamma} = (\gamma, \omega_\gamma)$ for Potts or SoS configurations

Let $\omega^0 \in \Omega$ be a fixed reference configuration. A **contour** for the general spin configuration $\omega \in \Omega$ **relative to** ω^0 is a pair $\bar{\gamma} = (\gamma, \omega_\gamma)$ where the support $\gamma = \{i \in S : \omega_i \neq \omega_i^0\}$ is a **connected component** of the set of incorrect points for ω (with respect to ω^0), and $\omega_\gamma = (\omega_i)_{i \in \gamma}$.

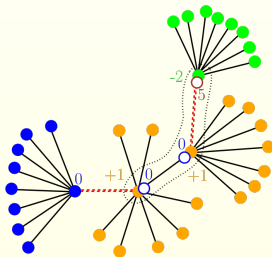


Figure: A contour $\bar{\gamma} = (\gamma, \omega_\gamma)$ relative to the GS ω^0 of Fig. 2.

Excess energy estimate for sparse ground states (general q -spin models)

Recall $u = \min_{k,j \in \mathbb{Z}_q} u_{k,l} > 0$ and denote $U = \max_{k,j \in \mathbb{Z}_q} u_{k,l} \geq u > 0$

Excess energy lemma

Let $\bar{\gamma}$ be a contour relative to ω^0 . Denote $\omega = (\omega_\gamma, \omega_{\gamma^c}^0)$ the corresponding excitation. Let $c := (d-1)u - d_D(U+u)$. Then

$$H(\omega) - H(\omega^0) \geq c |\gamma|$$

Note that $c > 0 \iff d_D < \frac{(d-1)u}{U+u}$ ($= \frac{d-1}{2}$ for Potts and Ising)

Corollary [2] : Stable ground states

For each interaction constants (u, U) , and each maximal internal degree $d_{\max} \in \mathbb{N}^*$, there is a minimal degree $d_{\max}(u, U, d)$ such that for $d \geq d(u, U, d_{\max})$ all the corresponding configurations ω^0 are **stable ground states**. We denote G^0 the set of stable ground set for the given model.

Peierls estimate and cluster expansion on trees

Let ω^0 be an inhomogeneous b.c. with partition function $Z_\Lambda^{\omega^0}$ at vol. $\Lambda \in \mathcal{S}$

Contour representation of the partition function

The partition function can be expanded into a contour representation

$$Z_\Lambda^{\omega^0} = \sum_{n \in \mathbb{N}} \sum_{\bar{\gamma}_1, \dots, \bar{\gamma}_n} \prod_{i=1}^n \rho[\omega^0](\bar{\gamma}_i)$$

where the sum is over pairwise compatible contours with activities

$$\rho[\omega^0](\bar{\gamma}) = \exp(-\beta(H_{\gamma \cup \partial\gamma}(\omega_{\gamma \cup \partial\gamma}) - H_{\gamma \cup \partial\gamma}(\omega_{\gamma \cup \partial\gamma}^0)))$$

On trees, **contours have no interior** (no symmetry needed !), implying :

Peierls bound for $\omega^0 \in \Omega$ at finite volume Λ

$$\mu_\Lambda^{\omega^0}(\sigma_j \neq \omega_j^0) = \sum_{\bar{\gamma}: \gamma \ni j} \rho[\omega^0](\bar{\gamma}) \cdot \frac{\sum_n \sum_{\bar{\gamma}_1, \dots, \bar{\gamma}_n \sim \bar{\gamma}} \prod_{i=1}^n \rho[\omega^0](\bar{\gamma}_i)}{\sum_n \sum_{\bar{\gamma}_1, \dots, \bar{\gamma}_n} \prod_{i=1}^n \rho[\omega^0](\bar{\gamma}_i)} \leq \sum_{\bar{\gamma}: \gamma \ni j} \rho[\omega^0](\bar{\gamma})$$

Results I : Inhomogeneous extremal states for q -spins and p -SoS and p -SoS

In both cases, there exists a finite $\beta_0 > 0$ s.t. for all $\beta \geq \beta_0$ there is a family of Gibbs measures $(\mu_\beta^{\omega^0})_{\omega^0 \in G^0}$ with the properties :

Theorem for q -spin and p -SoS models [1]

- ① *Identifiability* : $\omega^0 \neq \tau^0$ implies $\mu_\beta^{\omega^0} \neq \mu_\beta^{\tau^0}$
- ② *Extremality* : The measures $\mu_\beta^{\omega^0}$ are extremal in $\mathcal{G}(\gamma^{\beta\Phi})$
- ③ *Concentration* : $\mu_\beta^{\omega^0}$ concentrates around the ground states $\omega^0, \forall i \in S$,

$$\mu_\beta^{\omega^0}(\sigma_i = \omega_i^0) \geq 1 - Ce^{-c\beta}, C, c > 0 \text{ (} q\text{-spins case)}$$

$$\mu_\beta^{\omega^0}(\sigma_i - \omega_i^0 = k) \leq Ce^{-c\beta|k|^p}, \forall k \geq 0 \text{ (} p\text{-SoS case)}$$

- *Tools* : Cluster expansion, Fourier transform and cut-sets for extremality.
- Concentration property justifies the term "glassy" states used in [2,4].

Program

- 1 Introduction
- 2 Gibbs Measures on Lattice Spin Systems
- 3 Markov Fields on Trees, Markov chains and boundary laws
- 4 Uncountable Family of Inhomogeneous Extremal Gibbs States [1]
- 5 (Continuous) Extremal decomposition of Free States on Trees [2]

Previous results II: (non)-extremality of Ising and Potts free state – Reconstruction

Previous results for Ising free states

- **Higuchi [6]** : First remark that the 3^d Markov chain of Spitzer $\mu^\emptyset$ was not always extremal due to a tail-measurable fluctuating quantity.
- **Bleher-Ghanikodaev-Zagrebnev [7]** : Proof extremality/non-extremality at low/very low Temperature, same fluctuating likelihood ratios (effective fields)
- **Ioffe 96-97 [8]** : Simpler proof based on FK-percolation. Tail observable coincides with the EA overlap.

Remarks for Ising and Potts models

- Extremality/non-extremality is related to reconstruction on trees :
- Broadcasting on trees (Evans-Kenyon-Peres-Schulman 2002)
 - Reconstruction for Potts models (Sly 2014)
 - Disordered nature of the free Ising measure : Glassy states from [4].

Abstract expression of the decomposition measure – glassy states [2,4,5]

Boundary condition at infinity kernel [5] – glassy states [2,4]

Define for any ω , the tail-measurable kernel $\pi(\cdot|\omega) = \lim_{\Lambda \uparrow \mathcal{I}^d} \gamma_\Lambda(\cdot|\omega)$ if the limits exists, and $\pi(\cdot|\omega) := \nu$ arbitrary else.

Then, the limit exist and is an extremal Gibbs measure for μ -almost every ω .

DLR consistency and Backward martingale limit theorems implies

$$\forall \mu \in \mathcal{G}(\gamma), \mu(\cdot) = \int_{\Omega} \mu[\cdot|\mathcal{F}_{\infty}](\omega) d\mu(\omega) = \int_{\Omega} \pi(\cdot|\omega) d\mu(\omega)$$

Abstract formulation of the decomposition measure [5]

$$\forall \mu \in \mathcal{G}(\gamma), \mu = \int_{\text{ex}\mathcal{G}(\gamma)} \nu \cdot \alpha_{\mu}(d\nu)$$

where for all $M \in e(\text{ex}\mathcal{G}(\gamma))$, $\alpha_{\mu}(M) = \mu \left[\left\{ \omega \in \Omega : \pi(\cdot|\omega) \in M \right\} \right]$.

Continuity of the decomposition measure of the free states [2]

For the free state $\mu = \mu^\emptyset$, uncountably many glassy states play :

Theorem : Almost-sure singularity of extremals [2] at low T

$\exists \beta_0 > 0$, $\forall \beta > \beta_0$, for $\mu \otimes \mu$ -a.e. pair (ω, ω') , the extremal Gibbs measures $\pi(\cdot|\omega)$ and $\pi(\cdot|\omega')$ are singular w.r.t. each other, i.e.

$$\mu \otimes \mu \left(\{(\omega, \omega') \in \Omega \times \Omega : \pi(\cdot|\omega) \perp \pi(\cdot|\omega')\} \right) = 1.$$

$\implies$ the decomposition measure α_μ has no atoms and uncountably many extremal states enter in the extremal decomposition of $\mu^\emptyset$.

To see this last fact, assume $\mu(\pi(\cdot|\omega) = \mu_0) > 0$, for some atom μ_0 . Then

$$\begin{aligned} \mu \otimes \mu(\{(\omega, \omega') : \pi(\cdot|\omega) \perp \pi(\cdot|\omega')\}) &= 1 - \mu \otimes \mu(\{(\omega, \omega') : \pi(\cdot|\omega) = \pi(\cdot|\omega')\}) \\ &\leq 1 - \mu(\pi(\cdot|\omega) = \mu_0)^2 < 1. \end{aligned}$$

which is a contradiction.

Branch overlap and non-extremality of the free states [2]

Let $r = (r_1, r_2, \dots)$ be an increasing sequence of positive integers. Let $\Lambda_n = \Lambda_n^r = \{v_1, v_2, \dots, v_{n^2}\} \subset S$ so that $|\Lambda_n| = n^2$, and $|v_{i+1} - v_i| =: r_i$.

Thinned branch overlap

Define the tail-measurable *thinned branch-overlap*, measuring how much the configuration σ matches with ω on the increasing sparse volumes Λ_n , as

$$\underline{\phi}^\omega(\sigma) = \liminf_{n \uparrow \infty} \frac{1}{|\Lambda_n^r|} \sum_{v \in \Lambda_n^r} 1_{\sigma_v = \omega_v}.$$

Then, at low enough temperature, for sparse enough sets Λ_n ,

- ① $\exists \varepsilon(\beta) > 0$ s.t. for μ -a.e. ω , and $\pi(\cdot|\omega)$ -a.s. $\underline{\phi}^\omega \geq 1 - \varepsilon(\beta)$
- ② There exists a sequence of integers $r = (r_i)_i$ s.t. the strict inequality

$$1 \geq \pi(\underline{\phi}^\omega|\omega) > \pi(\underline{\phi}^\omega|\omega') \geq 0$$

holds for $\mu \otimes \mu$ -a.e. (ω, ω') .

Conclusions and Open Questions

Glassyness of free states is thus a generical phenomenon on trees

Extensions and works in Progress

- Extension to other hyperbolic graphs (Lobachevsky plane)
- Random graphs
- Disordered systems
- Random b.c. and spin-glass on trees (EA)
-

Other perspectives

- Reconstruction on trees and belief propagation algorithms
- Gradient models
- p -adic context
-

Main References

- [1] **L. Coquille, C. Külske, A. Le Ny.** Extremal Inhomogeneous Gibbs States for the SoS Model and Finite-Spins Models on Trees. *J. Stat. Phys.* **191**:71, december 2023.
- [2] **L. Coquille, C. Külske, A. Le Ny.** Continuity of the Extremal Decomposition of the Free State for Finite-Spin Models on Cayley Trees. *ArXiv Preprint*, submitted. arXiv-2310.1101, october 2023.
- [3] **D. Gandolfo, J. Ruiz, S. Shlosman.** A Manifold of Pure Gibbs States of the Ising Model on a Cayley Tree. *J. Stat. Phys.* **148**:999–1005, 2012.
- [4] **D. Gandolfo, C. Maes, J. Ruiz, S. Shlosman.** Glassy States : the Free Ising Model on a Tree. *J. Stat. Phys.* **180**:227–237, 2020.
- [5] **H.-O. Georgii.** Gibbs Measures and Phase Transitions. *de Gruyter*, 1988 & 2011.
- [6] **Y. Higuchi.** Remarks on the limiting Gibbs States on a $(d + 1)$ -tree. *Publ. RIMS Kyoto Univ.* **13**:333–348, 1977.
- [7] **P. Bleher, J. Ruiz, V. Zagrebnov.** On the Purity of the Limiting Gibbs State for the Ising Model on the Bethe Lattice. *J. Stat. Phys.* **79**, nos 1/2: 473–483, 1995.
- [8] **D. Ioffe.** On the Extremality of the Disordered State of the Ising Model on the Bethe Lattice. *Lett. Math. Phys.* **37**:137–143, 1996.